



Improving the Measurement of Financial Literacy by Modeling Noncorrect Responses

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Abstract: Very brief financial literacy tests are sometimes used in large surveys to reduce respondent burden, but when scored conventionally, they may appear too short to provide reliable or valid measurement. Using data from the Dutch LISS panel ($N = 4,860$), this study examines whether more fine-grained modeling of noncorrect responses provides meaningful information about financial literacy. Within an item response theory framework, we compare conventional binary scoring with alternative models that incorporate information from distractors and nonsubstantive response options. Particular attention is given to the nested logit model because it offers category collapsibility and preserves the substantive distinction between correct and noncorrect responses that is central to knowledge testing. Results show that this model yields higher reliability, particularly at low to average ability levels, and produces ability estimates that are more strongly associated with external validity criteria than conventional binary scoring. Our findings suggest that individual differences in financial literacy are reflected not only in whether answers are correct but also in how individuals respond under uncertainty, and that modeling this information can substantially improve measurement in short survey tests.

Keywords: financial literacy, item response theory, nested logit model, reliability, concurrent validity

Financial literacy, or the ability to understand and effectively use financial knowledge and skills, is an important individual difference linked to successful lifelong financial decision-making (Gignac, 2022; Gignac et al., 2023a,b). Sound economic decisions at the individual and household levels help avoid high-interest debt, grow income and savings through diversified investments, and promote long-term planning, while at the societal level they strengthen financial stability and reduce dependence on social safety nets (De Beckker et al., 2025; Gignac et al., 2023a). The OECD, G20, and the European Commission identify financial literacy as a vital life skill and emphasize the need to increase it worldwide given its relevance at the individual, household, and country-wide levels.

Due to its widely acknowledged importance, very brief financial literacy tests based on right/wrong multiple-choice items are often included in large-scale social surveys – for example, the four-item financial literacy scale used in the 2011 LISS panel (Dinkova et al., 2021; Kimhur, 2023) and Lusardi & Mitchell (2011) “Big Five,” five-item scale used in OECD surveys including the 2022 wave (with over 68,000 respondents from 39 countries) and the European Commission’s EU Financial literacy

survey (European Commission, 2023; OECD, 2023). The latter revealed that in the EU, only one in five individuals understands how bond prices relate to interest rates, and only 26% of EU adults possess a high level of financial knowledge (defined as answering four or five of the Big Five questions correctly).

A key challenge with very brief tests is that they can be unreliable and thus have low validity when conventionally scored (e.g., sum score, binary IRT score). In the present paper, we investigate whether alternative scoring strategies could be leveraged to improve measurement reliability and validity. Given the central role of right/wrong binary-scored financial literacy measures in influential surveys, an implicit but rarely examined assumption in conventional scoring is that all incorrect responses are equally uninformative. We argue that different noncorrect responses may actually reflect distinct levels of financial literacy. If so, conventional scoring may systematically discard information about individual differences in financial literacy. We examine this possibility by testing whether Item Response Theory (IRT) models that differentiate among incorrect responses yield additional meaningful information about financial literacy.

Beyond Correct versus Incorrect Responses

Financial literacy is commonly assessed with multiple-choice questions that include one correct answer, several incorrect options, and occasionally response categories such as “don’t know” or “would rather not say” (Dinkova et al., 2021; Lusardi & Mitchell, 2011). In most empirical work, responses are scored dichotomously, with only correct answers contributing to the total score and all other responses treated as equivalent (Dinkova et al., 2021; Kimhur, 2023), implying that noncorrect responses are uninformative regarding individual differences in financial literacy. However, this assumption may be overly restrictive, as distinct incorrect responses may carry informational value.

For example, previous research on cognitive ability tests suggests that some incorrect answers are “less wrong” than others in intelligence tests, and that selecting a comparatively less incorrect option can reflect a higher level of underlying ability (e.g., Myszkowski & Storme, 2018). A similar logic may apply to financial literacy tests. For instance, answering that the relationship between interest rates and bond prices is positive (incorrect) rather than stating that there is no relationship at all (also incorrect) may indicate partial knowledge of the existence of a relationship between interest rates and bond prices, even if the direction is misunderstood. Selecting the first incorrect option may therefore reflect a relatively higher level of financial literacy compared to selecting the second incorrect option.

Alternative response options such as “don’t know” or “would rather not say” may also convey meaningful information about financial literacy. Prior research in health and survey methodology indicates that respondents who select “don’t know” tend to possess lower levels of knowledge than those who provide substantive answers (Hay et al., 2015). Applied to financial literacy, choosing “don’t know” may similarly signal limited knowledge and thus lower levels of the underlying construct relative to respondents who attempt an answer, even if they are wrong. Although little research has directly examined the meaning of “would rather not say” responses, survey methodology studies suggest that such options belong to a broader class of nonsubstantive responses related to respondent engagement (Miller & Albaum, 2025). In financial literacy assessment, declining to answer may therefore reflect disengagement or avoidance of financial content, which could itself indicate low levels of financial literacy.

Recovering Information With Item Response Theory

If noncorrect responses contain information about financial literacy, then measurement models that differentiate

among them should yield more informative estimates of the construct. Item response theory (IRT) provides a useful framework for testing this idea. Standard dichotomous models, including one-parameter, two-parameter, and three-parameter logistic models, treat all noncorrect responses as equivalent and therefore ignore potentially meaningful variation in response behavior.

One way to incorporate such information is through the nominal response model (Bock, 1972), which allows each response category to relate differently to the latent trait. Under this approach, all response options – including distractors and alternative options such as “Don’t know” – can contribute distinct information about respondents’ levels of financial literacy. However, this high flexibility introduces two significant limitations. First, the model does not distinguish conceptually between correct and noncorrect responses, treating all options as unordered nominal choices. By treating all options as unordered nominal choices, the nominal response model allows the definition of the latent trait to be potentially influenced by specific patterns of incorrect guessing rather than actual correctness. Second, because the model calculates the probability of selecting any option relative to all others combined, it lacks category collapsibility. Modifying, removing, or merging a single distractor fundamentally alters the mathematical denominator for the remaining options. For example, if a popular distractor in a financial literacy question is deleted, the probability curve for the correct response artificially inflates, altering the item’s calculated difficulty and jeopardizing parameter stability.

To address these limitations, more structured approaches have been suggested, such as the nested logit model (Suh & Bolt, 2010), which preserves the correct-incorrect response distinction by modeling the response process in two stages: first distinguishing correct from noncorrect responses, and then differentiating among noncorrect responses (see Figure 1). The second stage is modeled using a nominal response structure, allowing distractors and alternative options to contribute differentially to the estimation of the latent trait while maintaining the interpretation of the latent variable as a knowledge-related ability. Such models can be viewed as a form of item response tree model (De Boeck & Partchev, 2012). By preserving the asymmetry between correct and noncorrect responses, nested logit models reduce the risk that latent trait estimates become driven by response categories that are only incidentally associated with ability. In addition, because all distractors are isolated inside an “incorrect option nest,” researchers can remove, add, or collapse choices within that nest without shifting the mathematical curve or parameters of the correct response. This category collapsibility ensures that specific

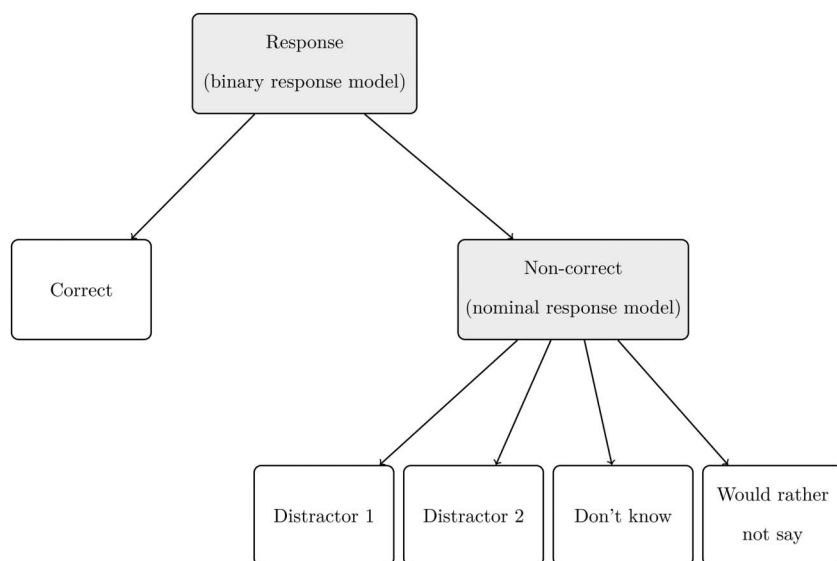


Figure 1. Conceptual representation of the nested logit model (example with 2 distractor options) used to model the full response process (shaded nodes represent the decision stages).

distractors can be modified or deleted over time while keeping the item's overarching difficulty and discrimination metrics stable.

Present Study

Research using nested logit and related models has shown that incorporating information from distractors improves the reliability of ability estimates, particularly at lower ability levels (Myszkowski & Storme, 2018; Storme et al., 2019). However, this literature has mainly focused on cognitive ability tests and on incorrect alternatives reflecting partial knowledge. Much less attention has been paid to the informational value of explicit nonresponse options such as “don't know” or “would rather not say.” Modeling such responses necessarily broadens the interpretation of the latent construct compared to standard dichotomous models. Rather than capturing only objective financial knowledge, nested logit-based scores may also reflect metacognitive aspects of financial literacy, such as how individuals manage uncertainty when they do not know an answer. If so, these estimates should correlate more strongly with self-reports of financial literacy than those derived from standard dichotomous models, including sum scores.

Whether this additional information would be reflective of financial literacy remains unclear. One way to address this question is to examine whether models incorporating noncorrect responses yield stronger associations with outcomes theoretically expected to be linked to financial literacy. If nested logit models capture information that is relevant in defining financial literacy, we can expect that its estimates will be more strongly associated with

objective financial behaviors that require and/or contribute to financial literacy – earning income from real estate, shares, or investment accounts – than standard dichotomous models (including sum scores).

This study examines this issue using a four-item financial literacy scale from the LISS panel. We compare standard dichotomous IRT models with models that explicitly account for incorrect and alternative response options, with a particular focus on the nested logit model. We evaluate whether the information retrieved from the complete list of response options improves measurement precision – as indicated by higher reliability – and measurement accuracy – as indicated by stronger associations with external criteria.

Method

Sample and Procedure

We analyzed data from 4,860 Dutch respondents (48.22% Male, $M_{\text{age}} = 49.69$, $SD_{\text{age}} = 14.93$). Participants were not forced to provide an answer. Across all items, only one participant provided a missing response. We excluded this participant from the analyses for simplicity. Data from the Longitudinal Internet Studies for the Social Sciences (LISS) panel (Scherpenzeel & Das, 2010) consist of a representative sample of Dutch individuals who participate in monthly internet surveys. The panel is based on a true probability sample of households drawn from the population register by Statistics Netherlands. Self-registration is not possible, and households that would otherwise be unable to participate are provided with a computer and

internet connection. The longitudinal LISS Core Study, consisting of separate topical surveys and core studies, runs every year since 2007 and covers a variety of domains including health, well-being, work, education, income, housing, leisure activities, political views, and personal values.

Measures

Financial Literacy

Objective financial literacy was measured using four single-choice questions from the LISS panel. The first item assesses understanding of interest compounding, asking respondents to evaluate the future value of a savings account with a fixed annual interest rate over several years. The second item assesses understanding of inflation and purchasing power, asking whether respondents would be able to buy more, the same, or less after one year when inflation exceeds the interest rate. The third item evaluates knowledge about risk and return, asking whether a single company share generally offers a more certain return than an equity investment fund. The final item assesses understanding of the inverse relationship between interest rates and bond prices, asking respondents to indicate how bond prices should change when interest rates increase. Each item has a single correct response, distractor(s), and alternative response options “Don’t know” and “Would rather not say.”

Validity Criteria

In the 2011 survey, the LISS panel also included a single-item measure of self-reported financial literacy. Respondents were asked to rate their understanding of financial matters on a seven-point Likert scale ranging from 1 (= *very poor*) to 7 (= *very good*).

We also examined concurrent validity using behavioral indicators derived from the Income core study of the LISS panel (Wave 4). Specifically, respondents reported whether they had generated income from 1) real estate, 2) shares, and 3) investment accounts. For each source of income, responses were dichotomized to indicate whether the respondent had generated any income from that source (0 = *no income generated*; 1 = *some income generated*). Note that the indicators we use should be interpreted cautiously, as the dataset does not provide information about the motivations or circumstances underlying these investment behaviors, they likely do not exclusively reflect financial literacy.

Data Analysis

Overview of Estimated Models

The psychometric modeling strategy consisted of first identifying a suitable measurement model for

dichotomous responses (i.e., correct vs. incorrect), and afterward extending this model to account for the full response structure of the items, using a nested logit model (Suh & Bolt, 2010). Because the nested logit model preserves the distinction between correct and noncorrect responses that is central to knowledge testing, it constituted the primary model of interest in the present study. As an additional comparison, we also estimated a nominal response model (Bock, 1972) to examine whether the results generalized to a more flexible categorical response framework. All models were successfully estimated using the expectation-maximization algorithm implemented in the R package *mirt* (Chalmers, 2012), using default parameters.

Regarding dichotomous models, we used common IRT models for binary data, including the one-parameter logistic model (1PL) and the two-parameter logistic model (2PL). Since the instrument has a non-negligible guessing probability due to its multiple-choice format, we also tested the three-parameter logistic model (3PL). In line with previous research (e.g., Dinkova et al., 2021), incorrect responses included altogether the distractors, “Don’t know,” and “Would rather not say.” Models were compared using a combination of likelihood ratio tests and deviance information criteria (AIC, BIC), which is common IRT practice.

The best fitting binary model (which was the 2PL, see results) was extended to a nested logit model – the 2-parameter nested logit model (2PNL) to account for the full categorical responses. This model separates the response process into two pseudo-items: 1) a correct vs. noncorrect response, and 2) conditional on a noncorrect response, a decision among distractors, “Don’t know,” and “Would rather not say.” Importantly, the underlying trait is assumed to influence both stages of the response process, allowing the noncorrect response options to provide additional information about respondents’ financial literacy.

Reliability and Validity Comparisons

For reliability comparisons, we used empirical reliability estimates from *mirt*, and bootstrapping (with 5,000 resamples) for inference on the binary versus categorical IRT reliability comparisons in line with prior research (Myszkowski & Storme, 2018; Storme et al., 2019). Empirical reliability estimates were computed using the *empirical_rxx()* function from the *mirt* package, based on expected a posteriori (EAP) factor score estimates and their associated posterior standard errors. Intuitively, this reliability estimate reflects the proportion of observed variance in the estimated latent scores that is attributable to true-score variance rather than measurement error. Computationally, the index is obtained as:

$$\text{Reliability} = 1 - \frac{\mathbb{E}(SE_{\theta}^2)}{\text{Var}(\hat{\theta})} \quad (1)$$

where $\mathbb{E}(SE_{\theta}^2)$ is the average posterior error variance across respondents and $\text{Var}(\hat{\theta})$ is the variance of the estimated latent scores. Thus, higher reliability indicates that the estimated scores contain relatively less measurement error.

To assess concurrent validity, standardized sum scores and factor scores (estimated using expected a posteriori methods) derived from each model were used to predict four external criteria: 1) self-reported financial literacy, and whether respondents reported having 2) income from real estate, 3) income from shares, and 4) income from investment accounts. Separate generalized linear models were estimated for each outcome, with standardized financial literacy scores entered as predictors. Model fit was compared using the Akaike Information Criterion (AIC), which allows comparison of models predicting the same outcome even when they are not nested. Lower AIC values indicate a stronger association between the specific financial literacy score and the criterion variable.

Because self-reported financial literacy was continuous, differences in convergent validity were also evaluated by comparing the correlations between each set of factor scores and the criterion using Steiger's (1980) test for dependent correlations, as implemented in the R package *cocor* (Diedenhofen & Musch, 2015). The difference in correlations was used as a measure of the difference in effect size.

Results

Selecting the Best Fitting Binary Response Model

The 2PL model (AIC = 18,676.89, BIC = 18,728.80) provided a substantially better fit than the 1PL model (AIC = 19,005.78, BIC = 19,031.73), as indicated by the likelihood-ratio test, $\chi^2(4) = 336.88$, $p < .001$. In contrast, the 3PL model (AIC = 18,681.69, BIC = 18,759.56) did not significantly improve model fit relative to the 2PL model, $\chi^2(4) = 3.20$, $p = .525$. The 2PL model was therefore retained as the best-fitting binary response model and subsequently extended to a nested logit specification.

As a side note, the absolute fit of the 2PL model was evaluated using the M2 statistic (Maydeu-Olivares & Joe, 2006). Despite a significant M2 test $M2(2) = 42.27$, $p < .001$, fit indices indicated acceptable model fit (RMSEA = .064, SRMR = .028, CFI = .977, TLI = .931). Comparable fit indices could not be computed for the 3PL or nested logit

models (due to too few degrees of freedom), and M2 is therefore reported only as an absolute fit check for the best fitting binary model. To assess the stability of the nested logit solution, the model was re-estimated 10 times using perturbed starting values generated by adding random noise ($SD = 0.3$) to the initial parameters. All replications converged to the same log-likelihood value, suggesting that the solution was stable and not driven by local maxima.

Gains from Modeling Noncorrect Answers

Reliability

As hypothesized, the 2-parameter nested logit (2PNL) model showed higher reliability ($r_{xx'} = .682$) than the baseline 2PL model ($r_{xx'} = .575$). This difference was statistically significant based on bootstrapping ($\Delta r_{xx'} = .106$, 95% bootstrapped CI [.100, .113]). A comparison of reliabilities as a function of ability level is shown in Figure 2 and indicates that, consistent with previous comparisons in the domain of intelligence (Myszkowski & Storme, 2018; Storme et al., 2019), the largest gains are obtained at low to average abilities. Figure 3 further illustrates how these reliability gains translate into differences in individual ability estimates by comparing EAP scores from the 2PL and 2PNL models. Differences are modest overall yet more pronounced at lower ability levels, consistent with the reliability patterns.

The 2PNL model item category curves are presented in Figure 4, and indicate that, beyond correct versus noncorrect responding, the alternative response options provide additional information about respondents' financial literacy. More specifically, the "Would rather not say" option is most prevalent at the lowest levels of the latent trait, indicating minimal knowledge or disengagement with the item content, whereas the "Don't know" option peaks at slightly higher trait levels, consistent with recognized uncertainty rather than complete lack of knowledge. Distractor options dominate intermediate regions of the trait continuum, reflecting partial or incorrect knowledge, while the probability of selecting the correct response increases monotonically with financial literacy.

Concurrent Validity

Financial literacy scores (standardized sum score, 2PL and 2PNL factor scores) were all three positively associated with the likelihood of reporting income from real estate, shares, investment accounts, and self-reported financial literacy (see regression coefficients in Table 1). However, scores from the nested logit model were better predictors of these variables than the standardized sum score and the 2PL factor score for all four criteria, with associated AIC

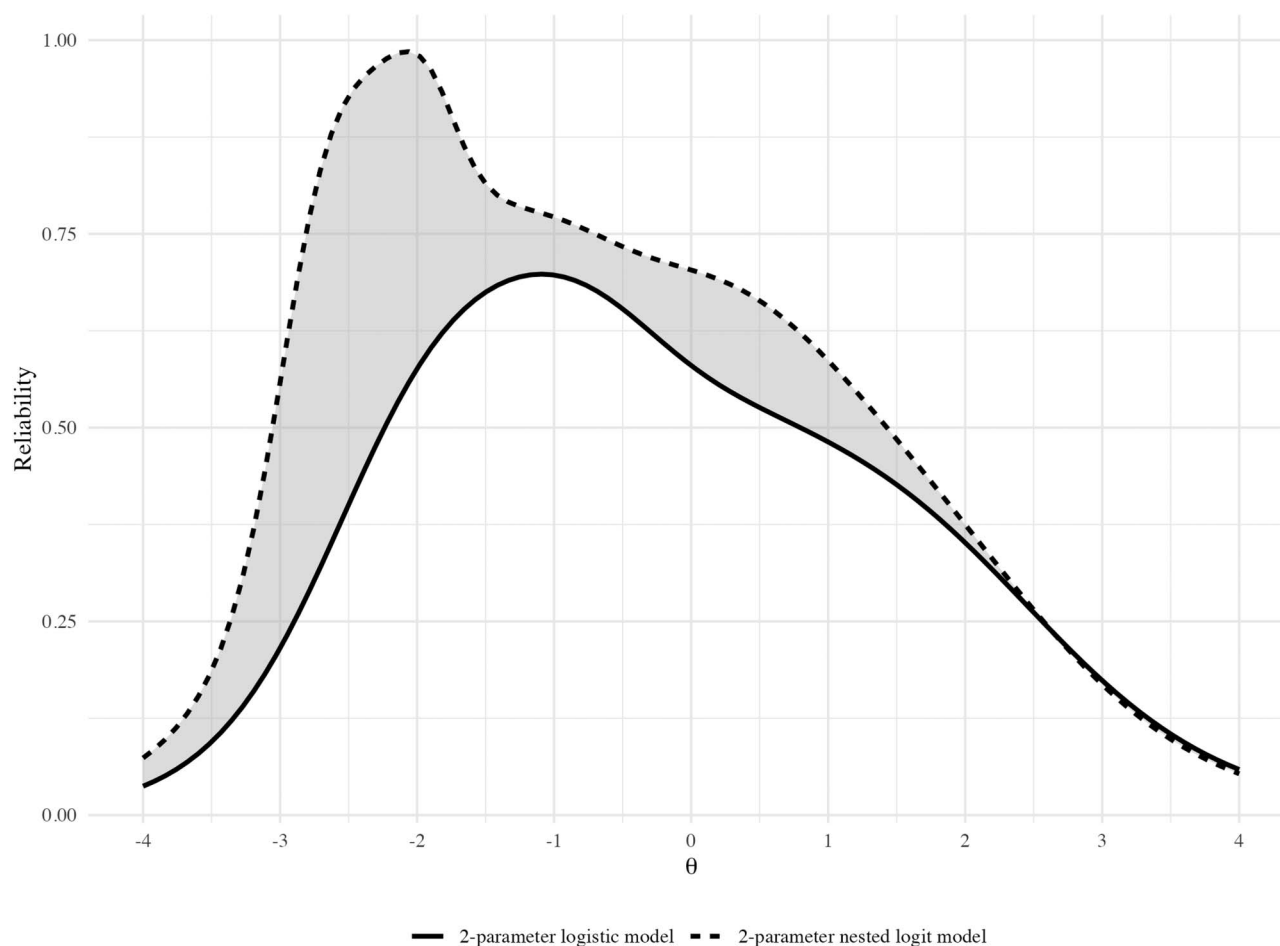


Figure 2. Overlaid reliability functions of the 2-parameter logistic and 2-parameter nested logit models.

values lower than those of the other models (see information criteria in Table 1).

For self-reported financial literacy, the sum score was moderately correlated with the criterion ($r = .222$), as were ability estimates derived from the 2PL model ($r = .216$) and the 2-parameter nested logit (2PNL) model ($r = .251$). The Steiger test for dependent correlations indicated that the sum score was slightly more strongly associated with self-reported financial literacy than the 2PL-based ability estimates, $z = 3.71$, $p < .001$. In contrast, the association was significantly stronger for the 2PNL-based ability estimates than for the 2PL-based estimates, $z = -5.50$, $p < .001$. Thus, while moving from sum scores to a conventional 2PL model did not improve convergent validity, explicitly modeling non-correct responses using a nested logit specification yielded a clear and statistically significant gain.

Comparison With the Nominal Response Model

To compare the nested logit approach with a more flexible alternative, we estimated a nominal response model

(NRM), which also incorporates information from distractor and nonsubstantive response options but does not preserve the distinction between correct and noncorrect responses in the definition of the latent trait. As with the 2-parameter nested logit (2PNL) model, the NRM yielded substantially higher reliability ($r_{xx'} = .669$) than the binary 2PL model ($r_{xx'} = .575$), although its reliability remained slightly below that of the 2PNL model ($r_{xx'} = .682$).

The NRM likewise improved prediction of the external validity criteria relative to the binary 2PL model, yielding lower AIC values for self-reported financial literacy (AIC = 13,478.99), income from real estate (AIC = 832.02), income from shares (AIC = 2,121.83), and income from investment accounts (AIC = 4,965.88). However, these improvements were broadly comparable to those obtained with the 2PNL model. Across the four criterion variables, neither model showed a consistent advantage: the 2PNL model yielded slightly lower AIC values for self-reported financial literacy, income from real estate, and income from investment accounts, whereas the NRM performed slightly better only for income from shares.

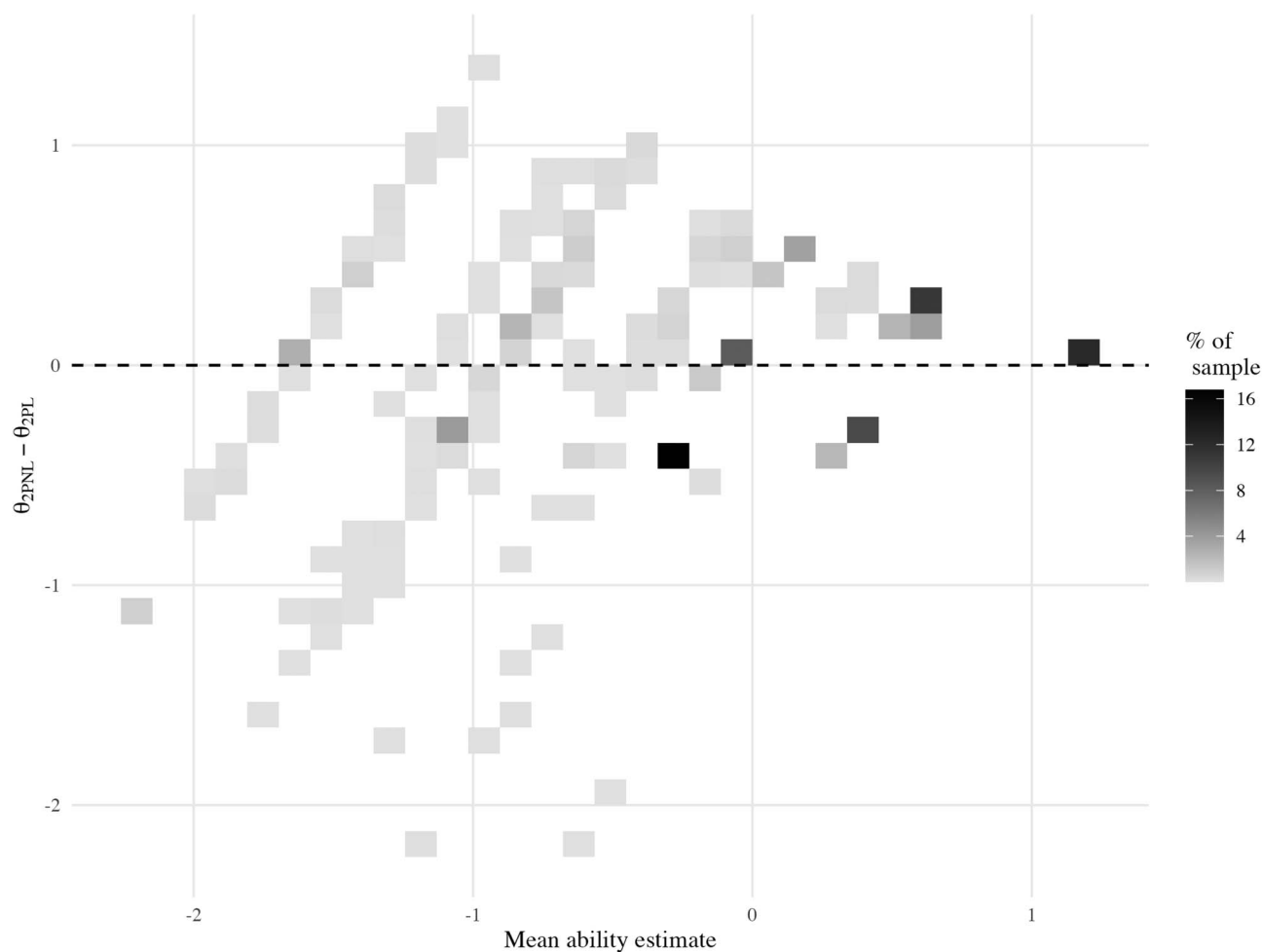


Figure 3. Bland–Altman plot comparing EAP ability estimates from the 2-parameter logistic (2PL) and 2-parameter nested logit (2PNL) models. Note. The plot is based on binned (hexagonal) counts to accommodate the discreteness of ability estimates derived from a small number of items. Shading indicates the percentage of respondents within each bin. The dashed horizontal line denotes zero difference between models.

As expected given its greater flexibility, the NRM provided a better fit to the response data (AIC = 33,167.02, BIC = 33,374.65) than the 2PNL model (AIC = 33,195.33, BIC = 33,402.97). However, this improved fit did not translate into consistently higher reliability or stronger prediction of the external validity criteria.

Discussion

In the present study, we examined whether meaningful information can be recovered from noncorrect responses in financial literacy assessment. Using data from a four-item financial literacy scale from the LISS panel, we first compared conventional binary IRT models (1PL, 2PL, 3PL) and found that the 2PL provided the best fit. We then extended this baseline model using a nested logit approach (Suh & Bolt, 2010) that incorporates the full response structure, including incorrect answers and alternative

options such as “don’t know” and “would rather not say.” Results showed that the nested logit model yielded higher reliability than the binary model, particularly at low to moderate levels of financial literacy, consistent with findings from intelligence research (Myszkowski & Storme, 2018; Storme et al., 2019). Extending this literature, we further showed that these reliability gains translated into stronger convergent validity, as latent scores from the nested logit model exhibited stronger associations with external financial criteria than those based on binary scoring. To our knowledge, this is the first study that empirically demonstrates that the nested logit approach can improve validity.

Theoretical Implications

Although no specific hypotheses were made about how noncorrect responses would vary across ability levels, clear

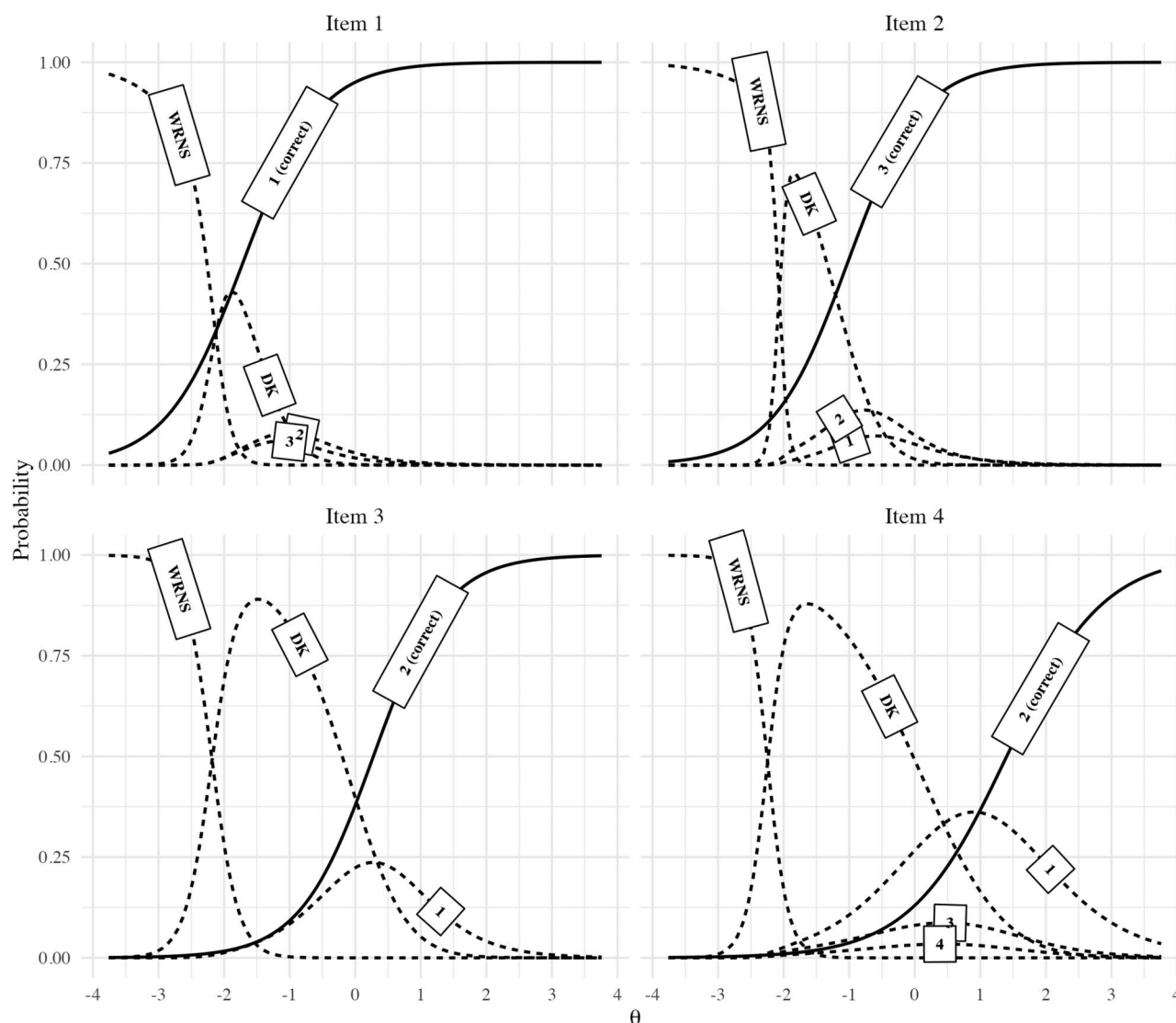


Figure 4. Item category curves from the 2-parameter nested logit model. Note. DK = “Don’t know”; WRNS = “Would rather not say.”

patterns emerged. Selecting “would rather not say” appears to characterize respondents at the lowest levels of the trait, consistent with disengagement, whereas “don’t know” is more common at slightly higher levels, reflecting acknowledged uncertainty rather than complete lack of interest. These results align with Lusardi and Mitchell (2023), who suggest that “don’t know” answers may reflect factors beyond missing knowledge. Recognizing one’s uncertainty – rather than disengaging – may reflect a form of epistemic awareness linked to financial literacy. This distinction is especially relevant in financial contexts, where awareness of one’s limits is central to sound decision making. Finally, individuals with higher financial literacy are more likely to attempt a substantive answer and to choose a comparatively “less wrong” option when incorrect. These patterns indicate that, for financial

literacy, noncorrect responses contain meaningful information lost under binary scoring.

Incorporating response options such as “don’t know” and “would rather not say” into the measurement model modestly but meaningfully broadens what the latent score represents. Rather than capturing only objective financial knowledge, the nested logit-based estimate also reflects how individuals manage uncertainty, incorporating aspects of metacognitive awareness (i.e., knowing what one does not know). This may explain why nested logit scores are stronger predictors of self-reported financial literacy than binary-scored estimates. Importantly, this broader interpretation does not appear to weaken construct validity: nested logit scores also show stronger associations with objective financial behaviors. In this sense, the approach may even enhance validity in domains where caution and

Table 1. Regression results (coefficients and AIC) for predicting validity criteria

	Scoring		
	Standardized Sum score	2-Parameter Logistic	2-Parameter Nested logit
<i>Predicting self-reported financial literacy</i>			
B (SE)	0.222 (0.014)	0.216 (0.014)	0.251 (0.014)
AIC	13,549.26	13,561.09	13,477.99
<i>Predicting income from real estate</i>			
B (SE)	0.754 (0.124)	0.759 (0.129)	0.839 (0.137)
AIC	835.11	836.43	831.30
<i>Predicting income from shares</i>			
B (SE)	0.995 (0.071)	1.007 (0.074)	1.125 (0.080)
AIC	2,152.93	2,159.91	2,129.11
<i>Predicting income from investment accounts</i>			
B (SE)	0.646 (0.038)	0.655 (0.039)	0.671 (0.039)
AIC	4,973.31	4,972.37	4,964.91

Note. B = regression coefficient; SE = Standard Error; AIC = Akaike Information Criterion.

self-assessment are integral to competent financial decision-making, such as saving, investing, or borrowing.

We additionally compared the nested logit model with the nominal response model (NRM), a more flexible model that also exploits information contained in noncorrect responses. The NRM expectedly yielded the best statistical fit due to its unconstrained flexibility, but this did not translate into superior measurement performance. Instead, the nested logit model (NLM) achieved slightly higher reliability and identical external validity. This represents a highly practical outcome: researchers can leverage the advantageous structural properties of the NLM – such as category collapsibility and independent parameter invariance – without sacrificing predictive power. However, the superior fit of the NRM also indicates that the cognitive processes underlying the choice of a response are likely more complex than the strict two-stage sequential hierarchy assumed by the NLM. Future research should evaluate more nuanced structures with the help of multidimensional nested logit models (Bolt et al., 2012), which maintain the correct-incorrect distinction but also captures secondary traits or systematic misconceptions within the distractor choices. Indeed, more complex IRTree models (De Boeck & Partchev, 2012) – which are increasingly applied in large-scale assessments (e.g., Myszkowski & Storme, 2025; Myszkowski & Storme, 2026) – could separate substantive knowledge from metacognitive or response-related tendencies such as uncertainty monitoring or disengagement. Recent work shows that nonsubstantive responses are not random but systematically related to individual differences and item characteristics (Storme et al., 2024). Applied to financial literacy, these models could test directly whether

responses such as “don’t know” or “would rather not say” reflect distinct latent traits beyond financial knowledge.

Practical Implications

The present findings closely mirror results from intelligence testing, where modeling information in noncorrect responses provides the greatest benefits for short tests and individuals at low to average ability levels (Myszkowski & Storme, 2018; Storme et al., 2019). In such contexts, limited test length constrains reliability, making the extraction of additional information from responses especially valuable. Our results suggest similar principles apply to financial literacy: Explicitly modeling structured variation among noncorrect responses can meaningfully improve both reliability and validity. As in intelligence research, the effectiveness of this approach depends on whether noncorrect responses reflect substantive processes rather than random error, underscoring the importance of theory-driven test design.

This is particularly relevant for the LISS panel, which relies on a four-item financial literacy scale. Such brevity may have been discouraging researchers from using the available data, as conventional sum scores are expected to yield low reliability. Our results confirm that this concern is warranted when responses are treated dichotomously. However, they also show that measurement quality can be substantially improved by modeling information contained in distractors and alternative response options. By recovering this otherwise discarded information, researchers can obtain more reliable and informative estimates of financial literacy. We hope that these findings will encourage greater use of the

LISS financial literacy data and, more broadly, illustrate how richer modeling of response processes can enhance the value of short instruments in large-scale surveys.

From a public policy perspective, financial literacy is a key determinant of individuals' financial well-being and economic security. Accordingly, policymakers invest in financial education programs aimed at improving financial knowledge and promoting sound financial behaviors. The present findings suggest that more refined measurement of financial literacy can improve the identification of individuals with low financial literacy, thereby enabling more effective targeting of those who are most likely to benefit from such interventions.

Conclusion

Financial literacy is reflected not only in whether an answer is correct but also in how individuals respond when they do not know the correct answer. Modeling these response patterns using a nested logit approach allows researchers to extract additional information from non-correct responses and obtain more precise and accurate estimates of financial literacy. This approach is particularly valuable when working with very short scales, where conventional scoring methods often suffer from limited reliability and, consequently, reduced validity.

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Conflict of Interest

The authors declare no conflicts of interest regarding this manuscript.

Publication Ethics

During the preparation of this work, the authors used ChatGPT to improve the language and readability of the manuscript. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Open Science

The information needed to reproduce all of the reported results is openly accessible. In the results section, we report exact *p*-values, effect sizes, and 95% confidence intervals when relevant.



Open Analytic Code: The R script is available on the Open Science Framework: <https://osf.io/mueds/> (Storme, 2026).



Open Data: Links to download the raw data on the LISS panel (Scherpenzeel & Das, 2010) website (<https://www.dataarchive.lissdata.nl/study-units/view/257>; <https://www.dataarchive.lissdata.nl/study-units/view/35>) are available at the beginning of the R script.



Open Materials: Links to download the study materials on the LISS panel (Scherpenzeel & Das, 2010) website (<https://www.lissdata.nl/>) are available at the beginning of the R script.



Preregistration of study and analysis plan: The study design and analysis plan were not preregistered. The sample size was determined by the number of observations that were available in the LISS data. In the sample subsection, we described how many observations were excluded and the reason why. Note that, since the analysis plan was not preregistered, the exclusion criterion (incomplete data for one participant) was not determined a priori.

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